A

Minor Project Report

On

NEO NOVA

Your Future AI Companion

Submitted in partial fulfillment of the requirements

For the award of the degree of

Bachelor of Technology

In

Computer Science and Engineering

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CERTIFICATE

This is to certify that the project report entitled " NEO NOVA " submitted by Mr. Aditya Sahani (CS-2341727), Mr. Mudit Agrawal (CS-2341718), Mr. Nikhil Raghav (CS-2341768), Mr. Parshant (CS-2341751) to IILM University, Greater Noida, Utter Pradesh in partial fulfillment for the award of Degree of Bachelor of Technology in Computer Science & Engineering is a bonafide record of the minor project work carried out by them under my supervision during the year 2023-2024.

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ACKNOWLEDGEMENT

We have taken efforts in this project, however, it would not have been possible without the kind support and help of many individuals and organizations. We would like to extend my sincere thanks to all of them.

We are highly indebted to Dr. Sreenu Banoth for his guidance and constant supervision. Also, we are highly thankful to them for providing necessary information regarding the project & also for their support in completing the project.

We are extremely indebted to Dr. Jasminder Kaur, HOD, ML & DS. We would also like to express our sincere thanks to all faculty and staff members of School of Computer Science and Engineering, for their support in completing this project on time.

We also express gratitude towards our parents for their kind co-operation and encouragement which helped me in completion of this project. Our thanks and appreciations also go to our friends in developing the project and all the people who have willingly helped me out with their abilities.

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ABSTRACT

This project presents NEO NOVA, a conversational AI chatbot developed using modern web technologies and cutting-edge AI tools to deliver dynamic and intelligent user interactions. The chatbot's user interface is built with HTML, CSS, and JavaScript to ensure accessibility, responsiveness, and an engaging visual design. At the core of NEO NOVA lies a robust natural language processing system powered by OpenAI, along with Google's Gemini API and Dialogflow ES/CX. These tools are used for understanding user inputs, generating context-aware responses, and managing multi-turn conversations.

The chatbot architecture utilizes Dialogflow to define intents, entities, and conversational flows, while the Gemini API enhances the system with advanced response generation and contextual understanding. Google API documentation and tools were used extensively for implementing language processing, voice recognition, and bot behavior refinement. The use of GPT-4 ensures that the chatbot remains flexible and capable of handling a wide range of user inputs with human-like fluency.

Unlike full-stack applications, this project does not incorporate user authentication, database connectivity, or complex backend services. Instead, NEO NOVA is focused purely on front-end deployment and API-driven intelligence, making it an ideal prototype for educational, demonstrative, or informational use cases. The modular architecture also allows future expansion toward more complex features if required.

In summary, NEO NOVA showcases how AI APIs and web development technologies can be combined to build a fully functional, interactive chatbot with natural conversation capabilities and real-time web deployment.

KEYWORDS: NEO NOVA, Open AI, Gemini API, Google APIs, Dialogflow, JavaScript, AI Web Application.

CONTENTS

CHAPTER 1: INTRODUCTION

In the rapidly advancing field of artificial intelligence, chatbots have emerged as one of the most practical implementations, significantly transforming user interactions across various digital platforms. This project introduces *NEO NOVA*, an intelligent, web-based chatbot developed using HTML, CSS, JavaScript, Google API, and Gemini API. The chatbot is designed to offer real-time, intelligent responses, making information access seamless and user-friendly. Unlike traditional systems that rely heavily on backend servers and complex user authentication processes, *NEO NOVA* maintains simplicity and efficiency by operating without user login, authentication, or third-party data storage. It functions entirely on the client side, leveraging API integrations to process user inputs dynamically. This architecture not only enhances accessibility but also ensures data privacy and low system overhead. The primary aim of the project is to demonstrate how modern AI APIs can be effectively utilized in web development to build responsive and context-aware systems that serve real-time communication needs.

CHAPTER 2: LITERATURE REVIEW / EXISTING WORK

Over the past decade, numerous chatbot systems have been developed using rule-based, retrieval-based, and generative AI approaches. Traditional chatbots primarily relied on predefined scripts or basic natural language processing, limiting their capability to understand complex queries. Notable advancements like Google Assistant, Siri, and Alexa showcase how AI-powered conversational agents have improved in contextual understanding and interaction. Most of these solutions, however, are integrated with user authentication systems and backend data processing, requiring robust infrastructure. Recent developments like OpenAI's GPT models and Google's Dialogflow platform have paved the way for more accessible, developer-friendly chatbot solutions. Prior works have demonstrated the feasibility of deploying AI-powered conversational interfaces in various domains, including customer service, education, and healthcare. However, many of these implementations are bound by limitations such as high operational costs, complex integrations, and data privacy concerns. Our review highlights the gap for a lightweight, client-side AI chatbot that can deliver intelligent interactions without requiring authentication or server-side handling—an aspect addressed by our project *NEO NOVA*.

CHAPTER 3: PROBLEM STATEMENT

Despite the widespread adoption of chatbots across industries, many existing solutions remain dependent on backend services, user authentication, and external data management. These dependencies often introduce challenges such as increased development time, higher hosting costs, and potential data privacy issues. Additionally, many chatbots require complex integrations, making them inaccessible for small-scale developers or educational projects. Users also face the inconvenience of having to log in or share personal data to interact with such systems. These factors limit the usability and scalability of chatbot technologies for lightweight applications. Therefore, there is a pressing need for a solution that allows the development of intelligent, interactive, and responsive chatbots that operate entirely on the client side. The problem this project addresses is: How can we design and implement a functional AI chatbot that delivers high-quality user interaction without relying on authentication, backend services, or third-party data handling? *NEO NOVA* is our response to this challenge—a lightweight yet powerful AI chatbot that leverages advanced APIs while maintaining a secure and simple architecture.

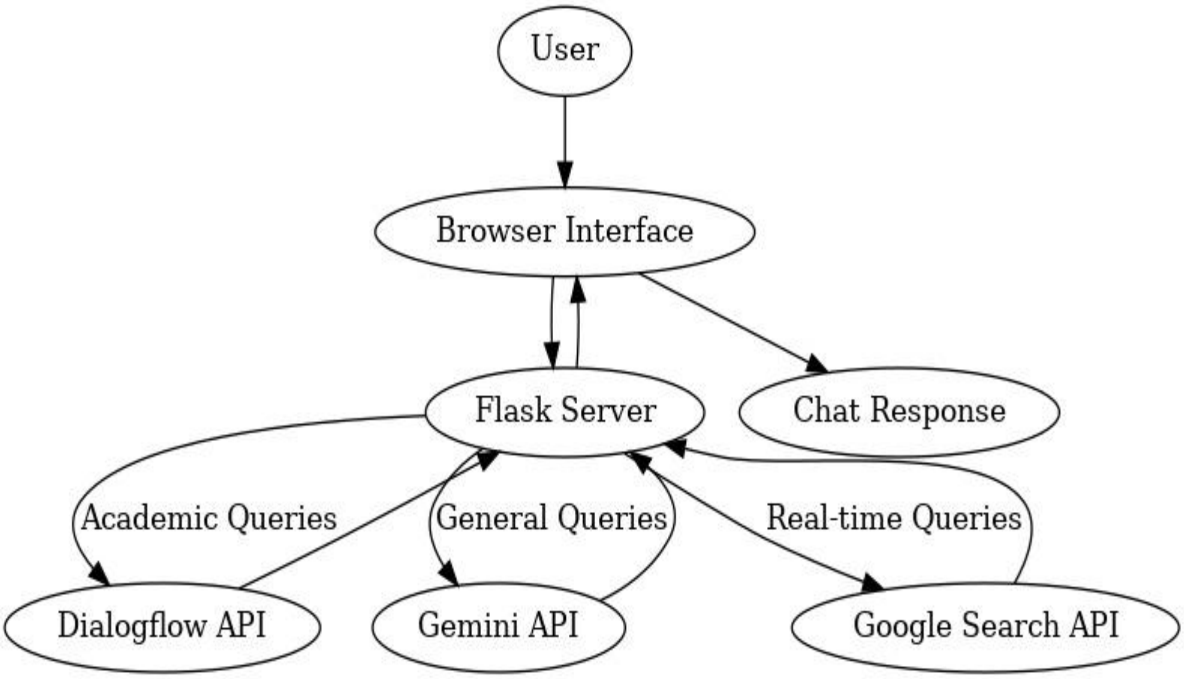
CHAPTER 4: PROPOSED WORK

The proposed project, *NEO NOVA*, aims to create a web-based AI chatbot that functions effectively without requiring user authentication or server-side processing. Our chatbot is built using a combination of frontend technologies—HTML, CSS, and JavaScript—alongside integrations with Google API and Gemini API to handle natural language understanding and response generation. The primary innovation lies in creating a completely client-side solution that performs all operations within the user's browser environment. This ensures real-time responsiveness, high accessibility, and enhanced user privacy. The chatbot interface is intuitive and designed for ease of use, allowing users to input queries and receive intelligent, context-aware replies. By using modern APIs, *NEO NOVA* can process complex inputs and deliver relevant outputs, simulating human-like conversation. The system architecture avoids reliance on databases or server calls, making it lightweight and ideal for demonstrations, educational tools, and limited-scale deployments. Overall, the project showcases how modern AI services can be leveraged to build practical, secure, and efficient chatbot solutions.

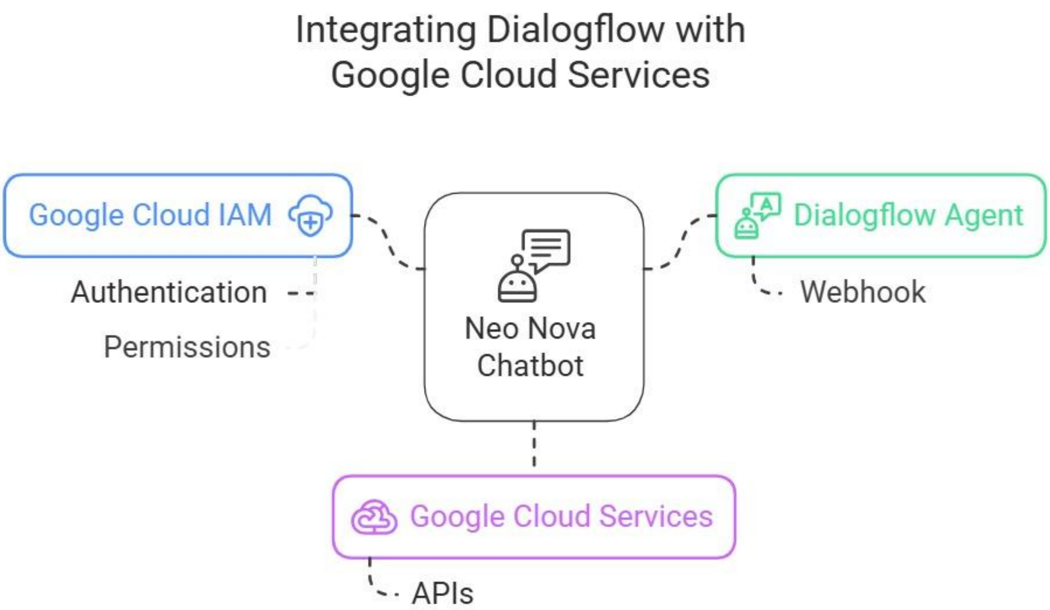
CHAPTER 5: SYSTEM DESIGN

The design of *NEO NOVA* follows a structured approach to ensure smooth interaction between the user interface and the AI processing components. A high-level flowchart outlines the chatbot's operation: the user inputs a message, the system sends the query to the Gemini API via JavaScript, receives a response, and displays it within the interface. The Data Flow Diagram (DFD) represents how data travels between modules: user input, API request, API response, and chatbot reply display. The Entity Relationship Diagram (ERD) is minimal since no database is involved, focusing instead on relationships between the frontend components and external APIs. The Use Case Diagram highlights user interactions with the system, emphasizing the simplicity of a no-login model where any user can ask questions and receive answers without authentication. These design elements ensure that *NEO NOVA* remains user-centric, lightweight, and functional, offering a seamless conversation experience through well-integrated web technologies and AI APIs.

Flow chart:



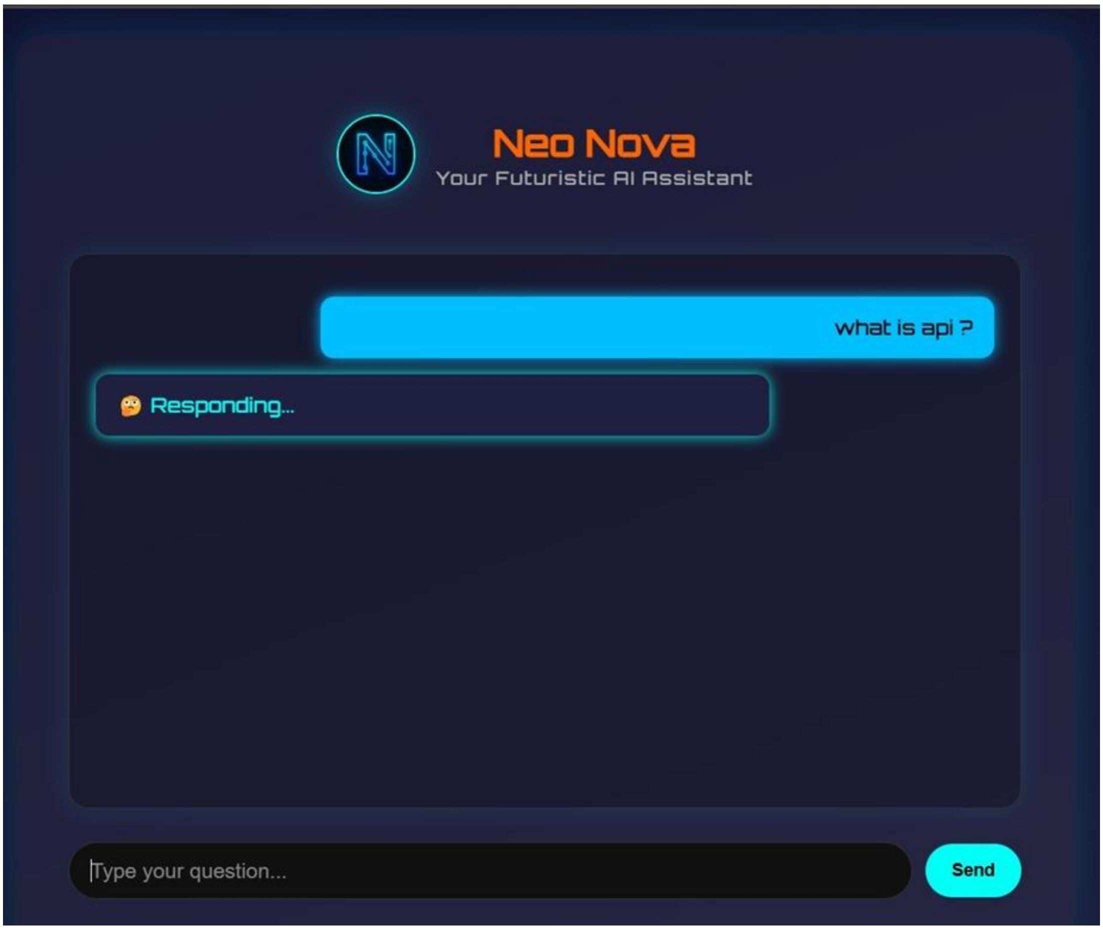
Integrating Dialogflow with Google Cloud Services :



CHAPTER 6: IMPLEMENTATION

Side technologies. The HTML structure forms the backbone of the chatbot interface, with CSS used to style the elements for a modern and engaging appearance. JavaScript is employed to manage all interactive elements, including capturing user input, displaying chatbot responses, and integrating with external APIs. The Google API and Gemini API are accessed through JavaScript fetch calls to handle user queries and generate intelligent replies. Each response is dynamically inserted into the chat window, maintaining a smooth conversational flow. The implementation also includes input sanitization to prevent errors and enhance user experience. Screenshots of the interface demonstrate a clean, intuitive chat layout with clear input and output sections. The chatbot operates efficiently across modern browsers without requiring any server-side hosting or database, making it easy to deploy and share. This implementation validates the feasibility of creating a robust, API-driven chatbot entirely on the client side.

Home View of Neo Nova :



CHAPTER 7: CONCLUSION, LIMITATION, AND FUTURE SCOPE

In conclusion, *NEO NOVA* successfully demonstrates the feasibility of developing a lightweight, AI-powered chatbot using modern web technologies and APIs without relying on backend infrastructure or user authentication. The project showcases the potential of Google API and Gemini API to create intelligent, real-time conversations on the web. However, limitations include dependency on internet connectivity for API access and limited customization in response tone or domain-specific knowledge due to the general-purpose nature of the APIs used. Additionally, since it's a purely client-side solution, advanced data tracking or personalized experiences are not feasible. For future scope, *NEO NOVA* can be enhanced with additional language support, integration with voice input/output, and domain-specific training for specialized applications. Expanding it into a modular architecture could also allow optional backend extensions for advanced features while preserving the current lightweight design. Ultimately, this project provides a scalable foundation for further innovation in AI-powered web communication tools.

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